

Rehire Likelihood Estimation

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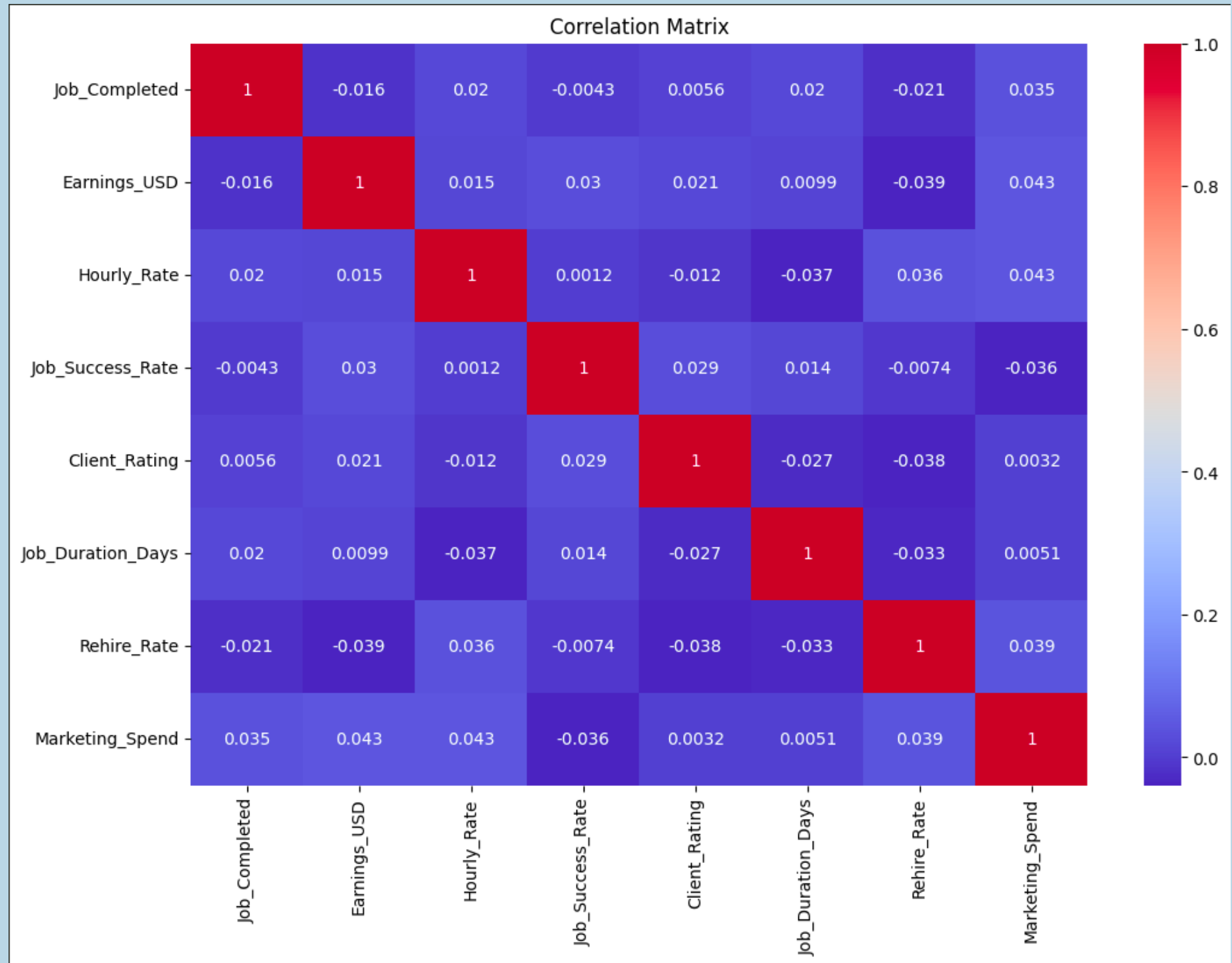


Problem

As digital plaforms scale with workforce, key metrics like rehire rate is substantial to optimiz-ing platform. In this work, I try to implement a machine learning framework for predicting free-lancer rehire rate using tabular data.

Problem & Dataset

The dataset consists of 1,950 structured records, each representing a unique freelancer on a simulated online labor platform. Each row aggregates behavioral, economic, and opera-tional signals across a time window, mimicking platforms like Upwork or Fiverr. Categorical features (e.g., Platform, Client_Region, Experience_Level) are one-hot encoded, while numerical features (e.g., Hourly_Rate, Job_Completed, Client_Rating) are z-score normalized. We also include several engi-neered variables such as earnings efficiency and client-region risk priors. The target variable is **Rehire_Rate**, a continuous value in $[0,100]$, indicating the proportion of repeat hires. The final dataset is sparse and high-dimensional, optimized for supervised regression models.



Models

Linear Regression

A baseline model that assumes a linear relation-ship between features and the target. Useful for interpretability and detecting linear trends.

Decision Tree & Random Forest

Decision Trees split data based on feature thresholds to minimize error, while Random Forests combine multiple trees to reduce over-fitting and improve generalization.

Multilayer Perceptron (MLP)

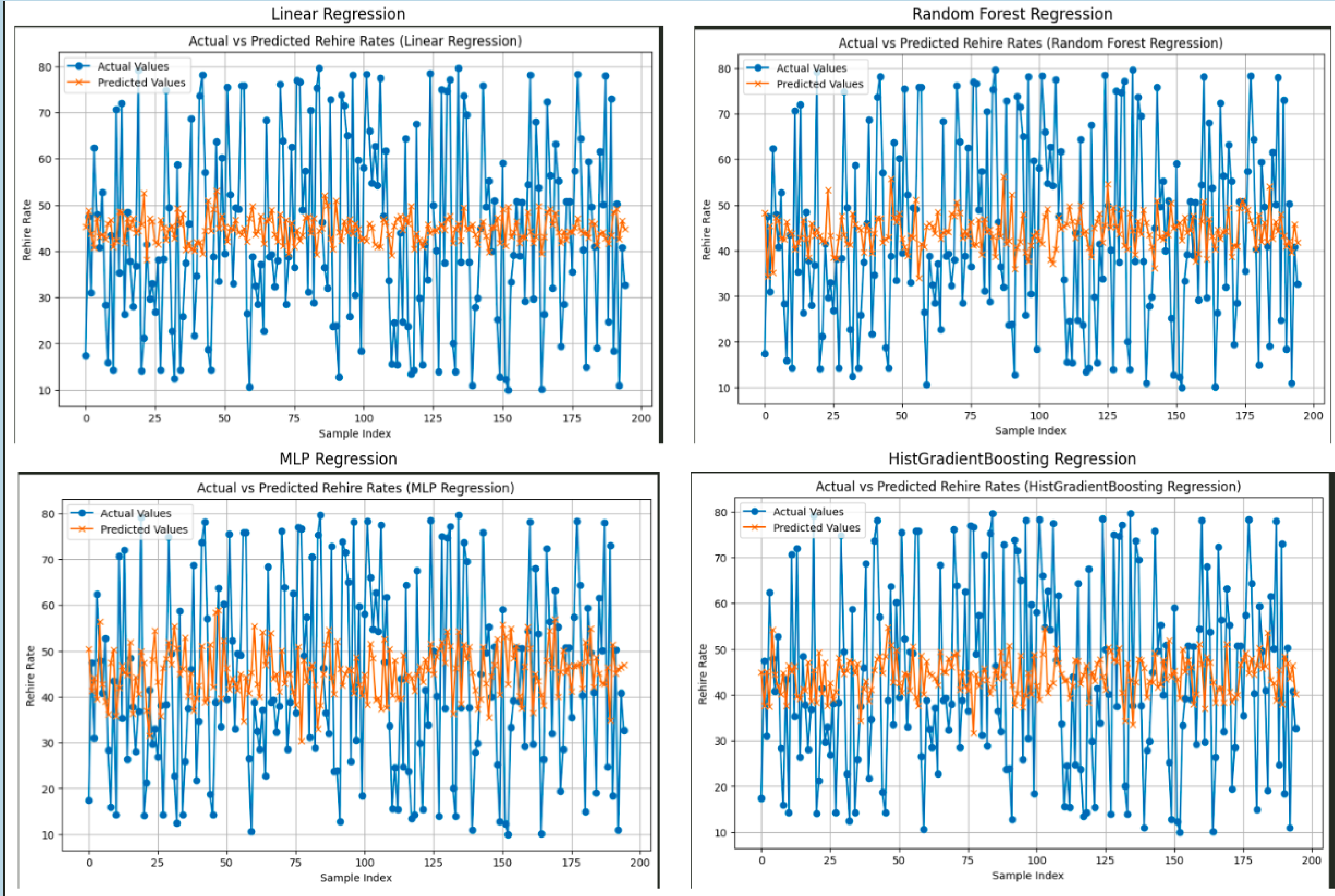
A feedforward neural network with one or more hidden layers, capable of modeling complex nonlinear relationships. Regularization and early stopping are applied to avoid overfitting.

Boosting Algorithms

We use gradient boosting algorithms (gradient boosting and histogram-based gradient boost-ing), which build models sequentially, where each new model corrects the errors of its prede-cessors. Effective for capturing subtle patterns in tabular data.

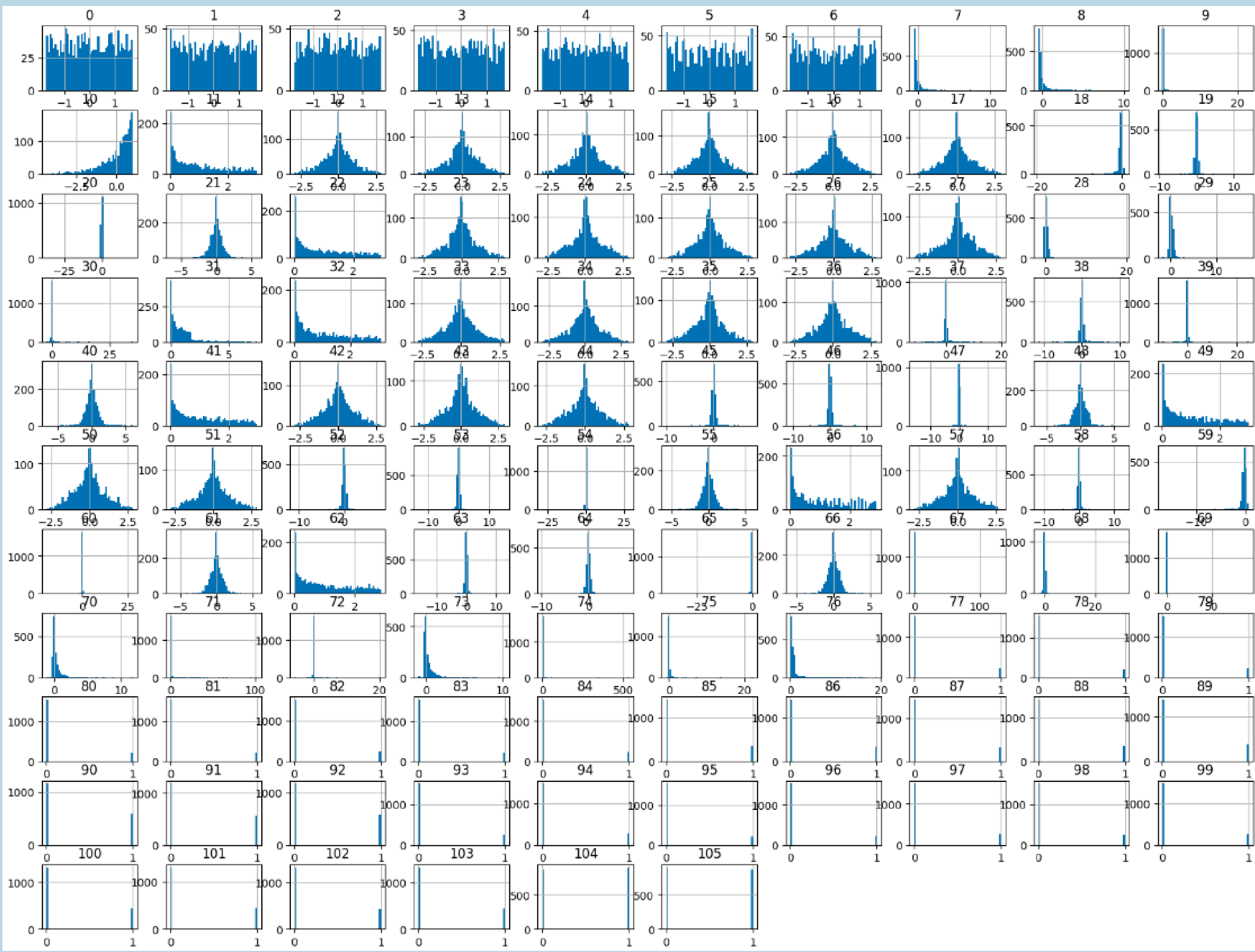
Experimental Result

We evaluated four regression models for predicting rehire rates: Linear Regression, Random Forest, MLP, and HistGradientBoosting. All models demonstrate similar prediction patterns, consistently forecasting values within a narrower range (approximately 40-55%) compared to the actual values which show significant volatility (10-80%). This consistent trend across all models suggests the limitation may be inherent to the dataset rather than the modeling approach. Among the tested methods, Random Forest and HistGradientBoosting performed marginally better at tracking extreme values, with Random Forest achieving the lowest Mean Absolute Error. Future work will focus on feature engineering to better capture the extreme values present in the rehire rate data.



Potential Improvements

Regression model performance is exceptionally sensitive to the underlying data characteristics and feature selection. **The same regression algorithm may produce significantly different pre-dictions depending on feature engineering and hyperparameter tuning.** The gap between predicted and actual values suggests opportunities for feature engineering and synthetic data gener-ating. The following charts illustrate distribution of features after feature engineering.



Addressing the limited prediction range could involve techniques such as data augmentation and targeted sampling. Additionally, incorporating temporal features may capture the volatility patterns observed in the actual rehire rates.